

MATH 4340/ COSC 4340 – Numerical Methods for Differential Equations

HW 2: Due by midnight on Saturday February 3, 2024.

Submit to WyoCourses a zipped folder containing a Jupyter notebook (both ipynb and HTML files).

Handwritten work should be included as a PDF or typed into the notebook as LaTeX.

1. [50 points] Let $\mathbf{y} = [x, y, z]$ and consider the following system of equations called the Rossler system,

$$\begin{cases} \frac{dx}{dt} = f_1(\mathbf{y}) = -y - z \\ \frac{dy}{dt} = f_2(\mathbf{y}) = x + ay \\ \frac{dz}{dt} = f_3(\mathbf{y}) = b + z(x - c). \end{cases}$$

Here, constants $[a, b, c]$ can be varied to change the system's dynamical behavior. (See the Wikipedia page.)

- (A). [15 point] Let $[a, b, c] = [0.2, 0.2, 5.7]$ and numerically integrate the system for $t \in [0, 60]$ and initial condition $\mathbf{y}_0 = [x_0, y_0, z_0] = [1, 1, 1]$ using the Forward Euler Method with $N = 10,000$ time steps. Make 2 plots. The first should show $x(t)$, $y(t)$ and $z(t)$ versus time t . The second plot should use the Python function `mpl_toolkits.mplot3d.Axes3d()` to show the system's trajectory in 3D.
- (B). [10 point] Repeat Problem 1A but explore different a values: $a \in \{0, 0.1, 0.2, 0.3, 0.38\}$. Keep parameters b and c the same. The dynamics should become more chaotic as a increases.
- (C). [15 point] Repeat Problem 1A but instead use the explicit Trapezoidal Method,

$$\begin{aligned} \tilde{\mathbf{y}}_{i+1} &= \mathbf{y}_i + hf(t_i, \mathbf{y}_i) \\ \mathbf{y}_{i+1} &= \mathbf{y}_i + \frac{h}{2} [f(t_i, \mathbf{y}_i) + f(t_{i+1}, \tilde{\mathbf{y}}_{i+1})], \end{aligned}$$

where $\mathbf{y}_i$ and $\tilde{\mathbf{y}}_i$ denote vectors. Compare your results to those found in Problem 1A.

- (D). [10 point] Repeat Problem 1B but use the explicit Trapezoidal Method instead of the FE method. Compare the results to those for the FE method. How does the parameter a influence whether the two numerical methods produce similar results?

2. [50 points] Consider the third-order differential equation $y'''(t) - y'(t) = t$ with the initial condition $y(0) = y'(0) = y''(0) = 0$.

- (A). [10 point] Prove that $y(t) = \frac{1}{2}(e^t + e^{-t} - t^2) - 1$ solves the differential equation and this initial condition.
- (B). [10 point] Derive a system of equations that is equivalent to this third-order differential equation and find a Lipschitz constant. *Hint: Recall the example from class for a linear, third-order system.*
- (C). [10 point] By hand, use the Forward Euler method to approximate the values of $y(t)$, $y'(t)$, and $y''(t)$ for $t \in \{0, 0.25, 0.5, 0.75, 1\}$.
- (D). [10 point] Create a function that uses the forward Euler method to numerically solve this system over the domain $t \in [0, 1]$, and use your code to validate your answers Problem 2C. Make a plot that compares the true and predicted values of $y(t)$, $y'(t)$, and $y''(t)$.
- (E). [10 point] Let $E_i = \|\mathbf{y}(t_i) - \mathbf{y}_i\|$ denote the global error, where $\|\cdot\|$ denotes any vector norm. Use the function `loglog()` to plot E_1 versus time step h and E_N versus h and numerically estimate their convergence rates as $h \rightarrow 0$. *Hint: I suggest using the values $N \in \{4, 40, 400, 4000\}$.*